Dual-Timescale Quantum Reservoir Computing for Financial Volatility Forecasting

QuantumEdge - GIC 2026 Phase 3 | Declared QPU: IBM ibm_fez | Independent replication: ibm_marrakesh
Judith Sarjeant (PM/Domain), Shema Nourine (Technical Lead), Hasarindu Perera (Data/ML), Van Tien Nguyen (Team Lead)

Abstract - Financial volatility combines long memory, nonlinear clustering, and abrupt regime changes, making it a natural test for reservoir computing. We present a dual-timescale quantum reservoir computing (QRC) architecture for one-day-ahead S&P 500 realized-variance forecasting and calm-versus-turbulent regime detection. Two fixed transverse-field Ising reservoirs encode fast shocks and slow persistence; only the classical ridge readout is trained. On a locked 750-day test, the calibrated QRC achieves RMSE 7.532x10^-5 and QLIKE 0.3261, improving on HAR-RV by approximately 5.9% and 21.0%, respectively. A QRC-feature classifier identifies calm and turbulent states with 78.1% accuracy against a 64.7% majority baseline. On authentic Oxford-Man five-minute realized variance, QRC is practically tied with HAR-RV. The declared IBM Fez validation reported feature correlation 0.989 and MAE 0.0569; an independent Marrakesh replication with complete 39-observable artifacts achieved 0.9916 and 0.0460. The contribution is not a quantum-supremacy claim; it is a reproducible, ablation-driven characterization of where a small hardware-native QRC helps, where it does not, and what remains before production use.

1. Introduction and Track Framing

Volatility forecasting matters because risk limits, derivatives pricing, hedging, market making, and execution quality all depend on the expected distribution of future price variation. The task is difficult for the same reasons it is economically important: realized variance is persistent over several horizons, reacts nonlinearly to shocks, and shifts between calm and turbulent states. HAR-RV models the multi-horizon persistence directly (Corsi, 2009), while recurrent neural methods attempt to learn it from data. Quantum reservoir computing offers a different hybrid route: a fixed quantum dynamical system performs the nonlinear temporal transformation, and a lightweight classical readout learns the forecast (Fujii & Nakajima, 2017).

Track A asks whether quantum reservoir methods can deliver useful dynamic-system forecasts under realistic execution constraints. We therefore define a narrow, falsifiable problem: forecast next-day S&P 500 realized variance and classify whether the next observation belongs to a calm or turbulent state. The evaluation includes five strong classical comparators, authentic intraday realized-variance validation, ablation, scaling, finite-shot and noise studies, and targeted execution on IBM hardware. The design follows recent QRC work on realized volatility while adding explicit fast and slow reservoirs, feedback memory, hardware-native readout selection, and a reproducibility package (Li et al., 2026).

The study makes four contributions. First, it maps the daily/weekly/monthly structure of volatility to two quantum reservoirs with deliberately different dynamical regimes. Second, it lets ablation determine the deployable architecture: entanglement-spectrum features were removed because they did not improve performance and were simulator-only. Third, it separates a reproducible Garman-Klass headline target from a genuine five-minute Oxford-Man validation target. Fourth, it validates observable-level transfer to the declared IBM platform rather than presenting a statistically meaningless three-window forecast correlation.

2. Dual-Timescale QRC Architecture

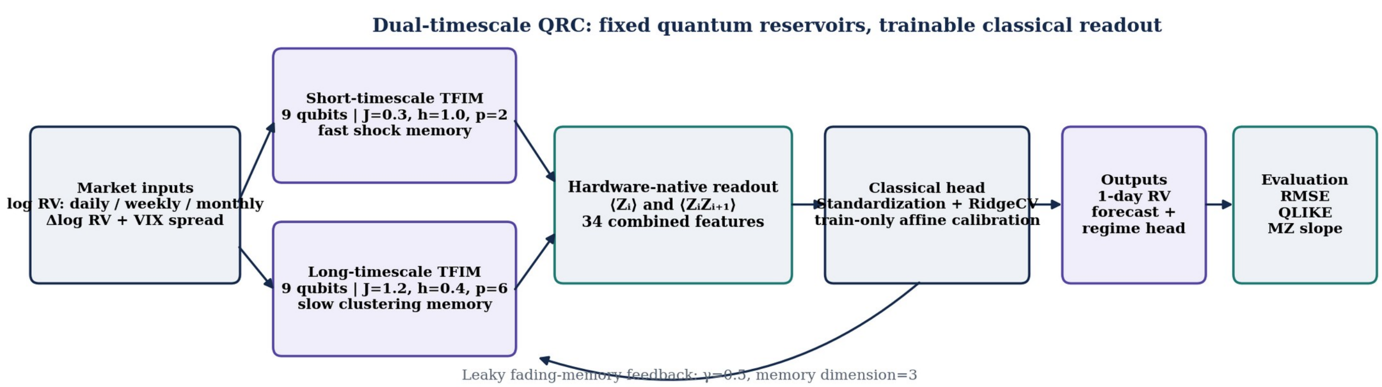

Figure 1. Dual-timescale QRC pipeline. Quantum reservoirs are fixed; training occurs only in the classical readout and optional affine calibration.

2.1 Reservoir Dynamics, Readout, and Memory

At time t, the five-dimensional input vector contains log realized variance at daily, weekly, and monthly horizons, the one-day change in log realized variance, and a VIX-implied-volatility spread. Inputs are scaled using training data and encoded as repeated RY rotations. Each reservoir applies nearest-neighbour RZZ interactions followed by RX rotations. The short reservoir uses nine qubits with J=0.3, h=1.0, and p=2 Trotter-style repetitions, producing a circuit depth of 14 in the reference construction. The long reservoir uses nine qubits with J=1.2, h=0.4, and p=6, with depth 30. These are not trained quantum parameters; they define complementary fixed dynamical transformations.

For each reservoir, the model measures the nine single-qubit expectations <Z_i> and eight nearest-neighbour correlations <Z_i Z_(i+1)>. Concatenating the two reservoirs yields 34 hardware-measurable features. A three-dimensional fading-memory state is calculated from the mean single-qubit response, mean pair response, and dispersion of single-qubit responses. The state is updated as m_t=(1-gamma)m_(t-1)+gamma s_t with gamma=0.5 and re-entered into the next encoding. The resulting recurrence adds memory without backpropagation through a quantum circuit.

A standardized RidgeCV head maps the quantum features to next-day log variance; the selected regularization coefficient is 0.3162. An optional affine calibration is estimated on training predictions only and then applied to the held-out test forecasts. This distinction is maintained throughout: the raw Dual+Pauli+FB model is the architectural result used in ablation, while the calibrated model is the best operational forecast. The two-state regime head is a separate logistic regression trained on the same QRC features, with the training-sample median realized variance defining the calm/turbulent threshold.

3. Theoretical and Analytical Justification

The architecture is designed around a signal-structure argument rather than a generic claim that quantum models are more expressive. Financial volatility exhibits heterogeneous persistence: daily shocks, weekly adjustment, and monthly clustering contribute differently to the next observation. HAR-RV captures this with explicit horizon averages (Corsi, 2009). QuantumEdge uses the same economic intuition at the feature-transformation layer: shallow, field-dominant dynamics respond quickly, while deeper, interaction-dominant dynamics retain slower structure. The classical ridge head is then free to combine both views.

The transverse-field Ising evolution introduces nonlinear interactions while remaining shallow enough for near-term execution. QRC avoids the repeated gradient optimization required by variational quantum neural networks; only the readout is trained, reducing exposure to barren-plateau and optimizer-instability problems. The model is therefore best understood as a structured nonlinear feature generator with memory. Any claimed benefit must survive comparison with ESN, the closest classical reservoir analogue, and HAR-RV, the strongest econometric benchmark for heterogeneous volatility persistence.

The ablation result is central to the theoretical claim. Entanglement-spectrum features appeared attractive in Phase 2 because they summarize global quantum structure, but they did not improve held-out accuracy. Removing them narrowed the model to Pauli observables that IBM hardware measures directly. The final claim is consequently stronger and more conservative: useful performance comes from the fixed dual-timescale dynamics and hardware-native observables, not from a simulator-only diagnostic.

4. Data and Experimental Methodology

Source	Variables	Role	Coverage / split
Yahoo Finance ^GSPC	OHLC, log return, Garman-Klass variance	Locked headline target and return series	3,773 days; 2010-01-05 to 2024-12-31
FRED VIXCLS / NFCI	VIX level; financial-conditions index	VIX spread feature; NFCI retained for provenance	Chronologically aligned to market file
Oxford-Man .SPX	Five-minute realized volatility	Authentic realized-variance validation	519 held-out observations

Table 1. Data strategy. The headline proxy is fully reproducible; the Oxford-Man sample tests transfer to authentic intraday realized variance.

After a 22-day feature window, 3,750 samples remain. The first 3,000 form the training set and the final 750 the untouched test set; no random shuffle is used. The target is one-day-ahead variance. Garman-Klass variance uses daily open, high, low, and close prices, while Oxford-Man supplies authentic intraday variance. Baselines are HAR-RV, ESN, early-stopped LSTM, GARCH(1,1), and persistence. Metrics include RMSE, QLIKE, Mincer-Zarnowitz slope, regime accuracy, and a secondary Sharpe measure. The final workflow also exports Diebold-Mariano comparisons, moving-block-bootstrap intervals, and transition-event metrics at +/-1 and +/-3 trading days. A separate mandatory MNIST notebook applies the same fixed-Ising, Pauli-readout design at 5, 10, and 15 qubits; its executed results are supplementary and are not used to support the financial ranking.

5. Results: Forecasting, Regimes, and Authentic Realized Variance

The calibrated QRC ranks first on the two primary forecasting criteria. Its RMSE of 7.532x10^-5 is 5.9% lower than HAR-RV and 5.8% lower than ESN. Its QLIKE of 0.3261 improves on HAR-RV by 21.0% and on ESN by 28.9%. The raw hardware-native QRC also leads both baselines, showing that the ranking is not created solely by calibration. GARCH and persistence are substantially weaker. LSTM is competitive but does not match QRC on RMSE or QLIKE.

Model	RMSE	QLIKE	MZ slope	3-state	Sharpe
QRC Dual+Pauli+FB (calibrated)	**7.532e-5**	**0.3261**	**1.465**	**57.1%**	**0.475**
QRC Dual+Pauli+FB	**7.725e-5**	**0.3794**	**1.298**	**61.7%**	**0.441**
ESN	7.993e-5	0.4588	1.101	59.1%	0.261
HAR-RV	8.005e-5	0.4129	1.262	59.3%	0.302
LSTM (early-stopped)	8.231e-5	0.4297	1.630	57.6%	0.499
Persistence	9.530e-5	0.6707	0.453	57.7%	0.041
GARCH(1,1)	1.010e-4	0.6656	-3.098	45.1%	0.329

Table 2. Locked 750-day test results. Lower RMSE and QLIKE are better; MZ slope closer to 1 indicates less conditional bias.

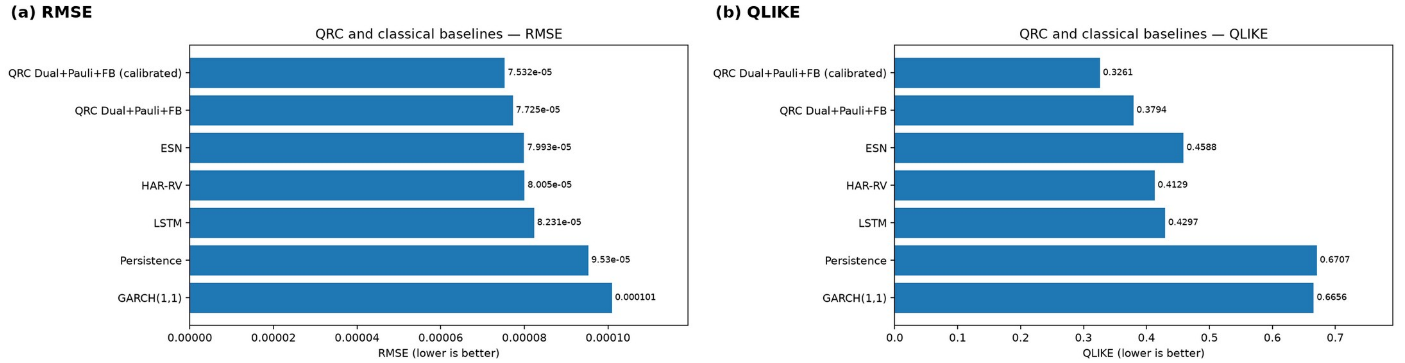

Figure 2. Headline baseline comparison. The calibrated QRC leads RMSE and QLIKE; the raw QRC also ranks first among uncalibrated models.

5.1 Bias and Secondary Metrics

The forecast ranking does not remove all bias. The raw QRC Mincer-Zarnowitz slope is 1.298 and the calibrated model's out-of-sample slope is 1.465; both exceed the ideal value of one. The joint Mincer-Zarnowitz tests reject intercept zero and slope one, so the paper reports residual under-response rather than describing the forecasts as unbiased. Likewise, Sharpe is not used to claim dominance: the early-stopped LSTM reaches 0.499, slightly above the calibrated QRC at 0.475. The raw QRC has stronger three-tertile agreement than its calibrated counterpart, illustrating that point-forecast calibration and categorical regime agreement need not improve together.

5.2 Calm/Turbulent Regime Detection

The dedicated two-state head is the more direct regime result. Using the training median as the threshold, QRC features support 78.1% calm-versus-turbulent accuracy, compared with 64.7% for always predicting the majority state. Because the use case concerns changes in liquidity and risk, the updated reproducibility notebook also detects true and predicted state changes, reports event precision/recall/F1, and measures daily accuracy inside +/-1-day and +/-3-day transition windows. These event-focused outputs prevent a high quiet-period score from being mistaken for early-warning performance.

5.3 Authentic Oxford-Man Validation

Model	RMSE	QLIKE	MZ slope
QRC Dual+Pauli+FB	**5.594e-5**	**0.2425**	**1.274**
HAR-RV	5.599e-5	0.2351	1.316
Persistence	6.067e-5	0.2612	0.672

Table 3. Authentic five-minute realized-variance validation (519-day test).

On genuine intraday realized variance, QRC and HAR-RV are effectively tied. QRC has the slightly lower RMSE (5.594x10^-5 versus 5.599x10^-5), while HAR-RV has the lower QLIKE (0.2351 versus 0.2425). Both beat persistence. The supported claim is therefore competitive performance against the strongest classical model on authentic realized variance, not statistical superiority. A paired Diebold-Mariano analysis is reserved for the next study.

6. Execution Studies: Ablation, Scaling, Shots, and Noise

Phase 3 requires more than a single benchmark table. The execution studies test which components matter and whether the selected model remains stable as hardware constraints are introduced. Together they support a robustness claim, but they do not show a scaling-based quantum advantage.

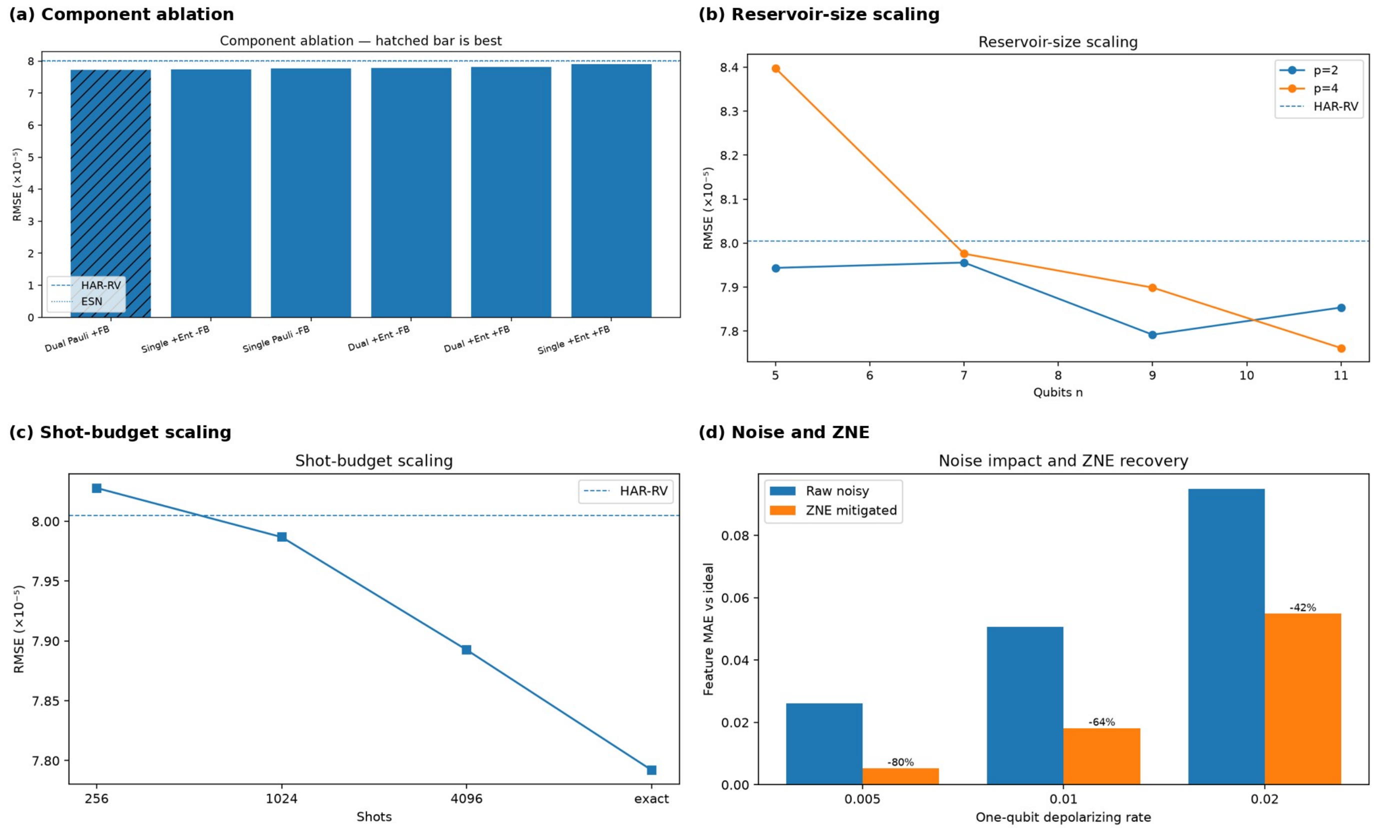

Figure 3. Execution studies. Dashed lines mark HAR-RV where applicable; lower RMSE or feature MAE is better.

6.1 Component Ablation

The best uncalibrated architecture is Dual Pauli+FB at RMSE 7.725x10^-5 and QLIKE 0.3794. Every tested QRC variant remains ahead of HAR-RV and ESN on RMSE, but the spread among QRC variants is small. Adding entanglement-spectrum features does not help; the best entanglement variant reaches 7.745x10^-5. The dual reservoir provides only a marginal improvement over single-reservoir alternatives, and feedback is approximately neutral across matched variants. These negative findings motivated the simpler, hardware-native Pauli model.

6.2 Reservoir Size and Encoding Density

Across n=5, 7, 9, and 11 qubits and p=2 or p=4 repetitions, RMSE stays in a narrow range, roughly 7.76x10^-5 to 8.40x10^-5. Most settings are at or below the HAR-RV reference, but there is no monotonic gain with qubit count. The p=4 setting is notably worse at n=5 and becomes competitive only at larger n. Encoding 3, 6, and 9 of the 9 available qubits produces RMSE 8.019x10^-5, 7.948x10^-5, and 7.792x10^-5, respectively, while QLIKE remains almost flat near 0.403-0.405. The result is robustness to design choice, not evidence that adding qubits automatically improves forecasting.

6.3 Finite-Shot Robustness

Shot sampling converges smoothly toward the exact state-vector result. RMSE falls from 8.028x10^-5 at 256 shots to 7.987x10^-5 at 1,024 shots, 7.893x10^-5 at 4,096 shots, and 7.792x10^-5 exactly. The 256-shot point is essentially at the HAR-RV line and the remaining finite-shot settings outperform it. QLIKE follows the same general convergence. The selected 4,096-shot budget is therefore conservative rather than essential for basic model viability.

6.4 Noise Injection and Zero-Noise Extrapolation

Depolarizing-noise experiments were run at one-qubit rates 0.005, 0.01, and 0.02, with two-qubit rates twice as large. Raw feature MAE rises from 0.0261 to 0.0950. Richardson-style zero-noise extrapolation recovers approximately 80%, 64%, and 42% of the injected-noise error. The final notebook adds the challenge-required amplitude-damping channel, composed with a 0.005/0.010 depolarizing background, and exports feature MAE for damping gamma 0.001-0.02. The interpretation remains conservative: mitigation helps at low noise but does not erase degradation at the highest tested rates.

7. Cross-Hardware IBM Validation

The simulator-first strategy reserves QPU time for the question hardware can answer credibly at this scale: does a real device reproduce the reservoir observables used by the classical readout? The declared Phase 3 backend is IBM ibm_fez. Its targeted seven-qubit circuit used optimization level 3, depth 68, 4,096 shots per circuit, and three validation windows. The locked Fez summary reported feature-level Pearson correlation 0.989 and mean absolute error 0.0569 across 39 observables. To test portability and recover complete observable-level evidence, the same three-window protocol was independently rerun on IBM ibm_marrakesh. All three Marrakesh jobs completed, at depth 71 and 4,096 shots, producing correlation 0.9916 and MAE 0.0460 over 39 paired simulator-hardware observations. The replication therefore matched, and slightly improved on, the declared-device agreement without changing the forecasting model.

Backend	Qubits	Depth	Shots	Windows / wall	Correlation	Feature MAE
ibm_fez (declared)	7	68	4,096	3 / ~211 s	0.9890	0.0569
ibm_marrakesh (replication)	7	71	4,096	3 / 488.36 s	0.9916	0.0460

Table 4. Cross-hardware QPU validation. Fez remains the declared Phase 3 platform; Marrakesh is an independent replication with complete observable-level artifacts.

A forecast-level correlation computed from only three windows would be unstable and potentially misleading, so it is deliberately excluded. Feature agreement is the relevant validation because these Pauli observables are the quantum outputs consumed by the readout. The full training and 750-day inference study remains simulator-assisted; the hardware experiments demonstrate transfer fidelity over targeted subsets rather than production-scale execution. The close agreement on two IBM Heron processors strengthens the portability claim, but it does not establish a runtime or scaling advantage.

8. Discussion, Stakeholder Impact, and Limitations

The evidence supports three practical conclusions. First, the dual-timescale QRC is a credible challenger for realized-variance forecasting: it leads RMSE and QLIKE on the locked headline sample and remains competitive with HAR-RV on authentic intraday data. Second, its features carry regime information beyond a majority rule, which is useful for risk desks that must change limits or hedging intensity as markets transition. Third, the final Pauli-only model is compatible with current gate-model hardware and shows high observable-level agreement on both the declared IBM Fez device and an independent IBM Marrakesh replication.

The evidence does not support a stronger claim. The headline edge is modest; Mincer-Zarnowitz slopes indicate residual bias; LSTM has the highest secondary Sharpe; authentic-RV performance is a tie with HAR-RV; entanglement-spectrum features do not help; and the tested qubit range shows no size advantage. Both hardware experiments are intentionally narrow. Results are also limited to the S&P 500 and one forecasting horizon. These limits matter because a small improvement can be operationally valuable in derivatives and risk management without constituting quantum supremacy.

For stakeholders, the near-term value is capability building and model diversification. A risk desk could use QRC as a challenger within an existing model-risk framework, compare its forecast and regime signal with HAR-RV and LSTM, and require stability, bias, and stress-testing thresholds before any capital or trading decision. The gradient-free reservoir reduces training complexity, but production adoption would still require monitoring, explainability of the classical head, data-lineage controls, drift tests, cost benchmarking, and fallback to approved classical models.

9. Reproducibility and Conclusion

The workflow is designed for judge reruns on qBraid. Notebook 1 reproduces financial results and exports pointwise predictions, transition-event metrics, Diebold-Mariano tests, blocked intervals, and both noise channels; Notebook 4 runs the common MNIST benchmark at 5/10/15 qubits; Notebook 2 rebuilds figures; Notebook 3 enforces package compliance. The repository includes a qBraid Skill, a Launch on qBraid README, deterministic seed 7, exact data hashes, cached or rebuildable Pauli features, complete IBM evidence, and a credential scan. The preserved core package achieved 27 PASS, 0 WARN, and 0 FAIL before the additional compliance checks were introduced.

QuantumEdge's Phase 3 result is an honest, reproducible demonstration of dual-timescale QRC for financial volatility. The model offers a measurable headline improvement, strong calm/turbulent classification, authentic-RV competitiveness, finite-shot and low-noise robustness, and repeatable feature agreement across two IBM Heron processors. The expanded evidence separates forecast ranking, event-focused usefulness, statistical uncertainty, common-benchmark expressivity, and hardware transfer. Until broader multi-asset and production tests are complete, the correct position is a hardware-ready forecasting challenger with cross-device replication, not a proven quantum advantage.

AI-use disclosure: Generative AI assisted with code review, debugging, notebook organization, figure preparation, document formatting, and drafting. The team selected the architecture and methods, executed the experiments, preserved and verified the evidence, interpreted the findings, and accepts responsibility for every submitted result and claim. No AI-generated numerical value is substituted for executed output.

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Submission assembly note: prepend the unmodified official GIC cover page and declare IBM ibm_fez as the primary Phase 3 platform. IBM ibm_marrakesh is an independent replication. The cover and references are excluded from the five-page body limit. The common MNIST, transition, statistical, and amplitude-damping outputs are provided in the executable supplementary notebooks and must be frozen before submission.